



Programming Fundamentals

LAB # 7

Course **Instructor Name:**

LAB Instructor:

Instructions:

- You have to submit CPP files of the program in a folder, named your Registration Number.
- You have to upload your lab tasks on CMS.
- All program codes should be written in C/C++. Students should use Visual studio compiler for coding.
- Indent and comment your code.
- Use meaningful variable names
- Plan your code carefully on a piece of paper before you implement it.

Learning Objectives:

- Demonstrate knowledge of **functions** programming C++.
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Write a program that reads a line of text (max 100 characters) from a file.

Count how many spaces are in the sentence.

Use the following functions:

1. `readFromFile()` – to read data from file into a C-string (by reference)
2. `countSpaces()` – takes a C-string and returns number of spaces
3. `displayCount()` – shows the result

Output:

Sentence read from file: Hello world, this is a file.

Number of spaces: 5

2.Instructions:

- A file contains 5 numbers.
- Your program should:
 1. Read the numbers from the file into an array.
 2. Calculate the average.
 3. Display the result.
- Use:
 - `readFile()` – reads values into array
 - `calculateAverage()` – returns average

o `displayAverage()` – displays the average

Output:

Values in file: 5 10 15 20 25

Average is: 15

3.Instructions:

- Read a sentence (max 100 characters) from the user.
- Count and display how many vowels it contains.
- Use:
 1. `getSentence()` – read the sentence (pass by reference)
 2. `countVowels()` – takes C-string, returns count of vowels
 3. `displayVowelCount()` – displays the result
- No I/O or logic in main. Use only C-strings.

Output:

Enter a sentence: Programming is fun!

Number of vowels: 5

4.Instructions:

- Store 10 integers in an array.
- Ask the user to enter a number to search.
- If found, show the index; else, show “Not Found.”
- Use:
 1. `inputArray()` – reads values into array (by reference)
 2. `searchIndex()` – receives array & number to search (returns index or -1)
 3. `displayResult()` – displays index or message

Output:

Enter 10 numbers: 12 43 2 9 34 56 3 78 21 0

Enter number to search: 56

Found at index: 5

or if not found:

Enter number to search: 100

Not Found

5.Instructions:

- Read 8 integers into an array.
- Divide array into two halves.
- Sum first and second half separately.
- Display each half's sum and the total.
- Use:
 1. `inputArray()` – read array
 2. `sumFirstHalf()` – returns sum of first 4
 3. `sumSecondHalf()` – returns sum of last 4
 4. `displaySums()` – shows both sums and total

Output:

Enter 8 numbers: 4 7 2 9 1 3 6 5

Sum of first half: 22

Sum of second half: 15

Total sum: 37

End of LAB 7

